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SACRIFICIAL LINER FOR COPPER INTERCONNECT

Abstract

A method and apparatus for forming an interconnect structure. The method includes depositing a ruthenium layer on a cobalt layer disposed within a feature formed on a substrate. The ruthenium layer has a ruthenium concentration that increases from a lower portion of the feature to an upper portion of the feature. The method includes depositing a copper layer within the feature. A material forming the copper layer is heated to a reflow temperature before, during, or after depositing the copper layer.

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Background/Summary

BACKGROUND
Field
[0001] Embodiments of this disclosure relate to an apparatus and a method of depositing layers on

a substrate.

Description of the Related Art

[0002] Integrated circuits may include more than one million micro-electronic field effect transistors (e.g., complementary metal-oxide-semiconductor (CMOS) field effect transistors) that are formed on a substrate (e.g., semiconductor wafer) and cooperate to perform various functions within the circuit. Reliably producing smaller features is one of the key technologies for the next generation of very large scale integration (VLSI) and ultra large-scale integration (ULSI) of semiconductor devices. However, as the limits of integrated circuit technology are pushed, the shrinking dimensions of interconnects in VLSI and ULSI technology have placed additional demands on processing capabilities. Reliable formation of the gate pattern is important to integrated circuits success and to the continued effort to increase circuit density and quality of individual substrates and dies.

[0003] Conventional methods of deposition have issues depositing materials in critical dimension (CD) openings associated with VLSI and ULSI technology (e.g., less than about 12 nm) due to pinch off of the deposited films. Resputter efficiency inside the structure is reduced during physical vapor deposition (PVD) processing, which often has an unacceptable thickness profile within the feature. Poor step coverage, overhang, and voids can be formed within features, such as vias or trenches, when the feature has a CD of less than about 12 nm. Insufficient deposition on the bottom and side walls of the vias or trenches can also result in deposition discontinuity, thereby resulting in device open circuitry or poor interconnection formation. Furthermore, the metal layer may have poor adhesion over the underlying material layer, resulting in peeling of the metal layer from the substrate and the subsequent conductive metal layer.

[0004] Therefore, there is a need for an apparatus and a method of depositing metal layers in small width critical dimension features with acceptable thickness profiles.

SUMMARY

[0005] The present disclosure generally relates to methods for forming an interconnect structure. In one embodiment, a method includes depositing a ruthenium layer on a cobalt layer disposed within a feature formed on a substrate. The ruthenium layer has a ruthenium concentration that increases from a lower portion of the feature to an upper portion of the feature. The method includes depositing a copper layer within the feature. A material forming the copper layer is heated to a reflow temperature before, during, or after depositing the copper layer.

[0006] In another embodiment, a method includes depositing a cobalt layer on a barrier layer, the barrier layer being disposed within a feature. The method includes depositing a ruthenium layer on the cobalt layer disposed within the feature. The ruthenium layer has a ruthenium concentration that substantially increases from a lower portion of the feature to an upper portion of the feature. The method includes depositing a copper layer within the feature. A material forming the copper layer is heated to a reflow temperature before, during, or after depositing the copper layer.

[0007] In yet another embodiment, a method includes depositing a ruthenium layer on a cobalt layer disposed within a feature. The ruthenium layer has a ruthenium concentration that increases from a lower portion of the feature to an upper portion of the feature. The method includes depositing a copper layer within the feature. A material forming the copper layer is heated to a reflow temperature before, during, or after depositing the copper layer. The method includes removing a top portion of the interconnect structure. The top portion of the interconnect structure comprises a first portion of the ruthenium layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments of the disclosure and are therefore not to be considered limiting of its scope, as the disclosure may admit to other equally effective embodiments.

[0009] FIG. **1**A illustrates a schematic side view of a processing chamber, according to at least one embodiment.

[0010] FIG. **1**B illustrates a schematic side view of a physical vapor deposition (PVD) processing chamber, according to at least one embodiment.

[0011] FIG. **2** illustrates a schematic top view of a multi-chamber processing system, according to at least one embodiment.

[0012] FIG. **3** is a flow diagram of method operations for depositing a plurality of layers, according to at least one embodiment.

[0013] FIGS. **4**A-**4**D illustrate a schematic side view of a device structure during various operations of the method of FIG. **3**, according to at least one embodiment.

[0014] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0015] Embodiments disclosed herein generally relate to methods of depositing a plurality of layers on a patterned substrate that contains features. A cobalt (Co) layer is deposited using a chemical vapor deposition (CVD) process in a plurality of feature definitions in a device structure formed on a substrate. In some embodiments of the method described herein, a ruthenium (Ru) layer is deposited over a cobalt (Co) containing layer that is formed in the plurality of feature definitions of the device structure using a physical layer deposition (PVD) process. One or more copper layers can then be deposited over the ruthenium layer, and the copper layers can be reflowed to create a copper layer that has a desired thickness profile. The Ru layer deposited over the Co layer may encourage increased flowability, reducing poor step coverage, overhang, and voids in the formed copper layer. The desired thickness profile increases device yield, reliability of the device structure, reducing device open circuitry and improving interconnection formation. Furthermore, thickness reduction associated with achieving the desired thickness profile can allow deposition processes to be extended to higher aspect ratio applications. Embodiments disclosed herein may be useful for, but are not limited to, increasing the flowability of deposited copper layers using dopants, such as Ru, Co.

[0016] As used herein, the term “about” refers to a +/−10% variation from the nominal value. It is to be understood that such a variation can be included in any value provided herein.

Processing System Examples

[0017] FIG. **1**A illustrates a schematic side view of a processing chamber **150**, according to at least one embodiment. The processing chamber **150** is configured to deposit a variety of materials, such as metal materials, onto interconnect structures **400** (FIG. **4**) formed on a substrate **153** disposed within the processing chamber. As shown, the processing chamber **150** includes a pedestal assembly **190**, a lid assembly **100**, a chamber body **120**, and a control unit **180**. The chamber body **120** is configured to protect substrates disposed within the body from the environment outside the body. Metal materials that can be deposited in the processing chamber **150** can include cobalt (Co), copper (Cu), nickel (Ni), ruthenium (Ru), derivatives thereof, or combinations thereof. The processing chamber **150** can be used to perform CVD, plasma enhanced-CVD (PE-CVD), pulsed-CVD, ALD, PE-ALD, derivatives thereof, or combinations thereof. The processing chamber **150** can also be used to anneal previously deposited metal layers. Thus, both the deposition processes and the subsequent annealing may be performed in-situ in the same processing chamber **150**. In other embodiments, deposition of material and annealing of material can be performed in separate chambers.

[0018] As shown, the lid assembly **100** includes a showerhead **156** with holes **109**, a blocker plate **140**, a water cooling cover plate **134**, a convolute liquid channel **162**, a gas box plate **160**, and a lid isolator **175**. The lid assembly **100** is configured to deliver gas through the holes **109** onto the substrate **153** disposed on a substrate supporting surface of a heater pedestal **152**, which is disposed below the showerhead **156**. Water channels, such as a convolute liquid channel **162**, are used to regulate the temperature of the lid assembly **100** during the vapor deposition process for depositing material. The lid assembly **100** is heated or maintained at a temperature within a range from about 100° C. to about 300° C., such as from about 125° C. to about 225° C., or from about 150° C. to about 200° C., according to at least one embodiment. The temperature can be maintained during the vapor deposition process of a cobalt-containing material, nickel containing material, or copper-containing material.

[0019] The showerhead **156** has a relatively short upwardly extending rim **158** coupled with the gas box plate **160**. Both the showerhead **156** and the gas box plate **160** can be formed from or contain a metal, such as aluminum, stainless steel, or alloys thereof. The convolute liquid channel **162** is formed in the top of the gas box plate **160** and covered and sealed by the water cooling cover plate **134**. Water is generally flown through the convolute liquid channel **162**. In one example, the lid assembly **100** is configured to be heated or maintained at a temperature of about 150° C. and is in fluid communication with a source of a cobalt precursor, such as dicobalt hexacarbonyl butylacetylene (C12H10O6Co) (CCTBA), and a source of a hydrogen precursor, such as hydrogen gas (H2).

[0020] The showerhead **156** has a plurality of apertures or holes **109** communicating between the lower cavity **130** and a processing region **126** to allow for the passage of processing gas. The processing gas is supplied through a gas port **132** formed at the center of the gas box plate **160** which is made of aluminum and is water-cooled. The upper side of the gas box plate **160** is covered by the water cooling cover plate **134** surrounding the upper portion of the gas box plate **160** that includes the gas port **132**. The gas port **132** supplies the processing gases to an upper cavity **138** which is separated from the lower cavity **130** by a blocker plate **140**. The blocker plate **140** has a large number of holes **109** disposed therethrough. In one implementation, the cavities **130** and **138**, showerhead **156**, and blocker plate **140** evenly distribute the processing gas over the upper face of the substrate **153**.

[0021] The pedestal assembly **190** is configured to support a substrate **153** that includes interconnect structures **400** during processing of the substrate. As shown, the pedestal assembly **190** includes a heater pedestal **152**, a pedestal stem **154**, lift pins **118**, a lifting ring **116**, and a lift tube **117**. The heater pedestal **152** is connected to the pedestal stem **154** that may be vertically moved within the processing chamber **150**. The heater portion of the heater pedestal **152** can be formed of a ceramic material. In its upper deposition position, the heater pedestal **152** holds a interconnect structures **400** in close opposition to a lower surface **107** of the showerhead **156**. A processing region **126** is defined between the heater pedestal **152** and the lower surface **107** of the showerhead **156**.

[0022] The substrate **153** is supported on the heater pedestal **152**, which is illustrated in a raised, deposition position. In a lowered, loading position, the lifting ring **116** is attached to the lift tube **117** which lifts four lift pins **118**. The lift pins **118** fit to slide into the heater pedestal **152** so that the lift pins **118** can receive the substrate **153** loaded into the chamber through a loadlock port **119** in the chamber body **120**. The heater pedestal **152** can additionally include a confinement ring **110** for plasma-enhanced vapor deposition processes.

[0023] A side purge gas source **123** can be coupled to the processing chamber **150** and configured to supply purge gas to an edge portion **151** of the interconnect structures **400** as needed. The purge gas can include H.sub.2, argon gas (Ar), nitrogen gas (N.sub.2), helium gas (He), combinations thereof, or the like. Furthermore, a bottom purge gas source **125** can also be coupled to the processing chamber **150** to supply the purge gas from the bottom of the processing chamber **150** to the substrate surface. Similarly, the purge gas supplied from the bottom purge gas source **125** can include a H.sub.2, Ar, N.sub.2, He, combinations thereof, or the like.

[0024] The lid isolator **175** is interposed between the showerhead **156** and a lid rim **166**, which can be lifted off the chamber body **120** to open the processing chamber **150** for maintenance access. The vacuum within the processing chamber **150** is maintained by a vacuum pump **170** connected to a pump plenum **172** within the processing chamber **150**, which connects to an annular pumping channel **174**.

[0025] A chamber liner **179** made of quartz is disposed in the processing chamber **150** which defines a side of the annular pumping channel **174**, but also partially defines a further choke aperture **181** disposed between the processing region **126** and the annular pumping channel **174**. The chamber liner **179** can have an annular shape. The chamber liner **179** also supports the confinement ring **110** in the lowered position of the heater pedestal **152**. The chamber liner **179** also surrounds a circumference at the back of the heater pedestal **152**. The chamber liner **179** rests on a narrow ledge in chamber body **120**, but there is little other contact, so as to minimize thermal transport. Below the chamber liner **179** is located a lower chamber shield **121**, made of opaque quartz. The lower chamber shield **121** can be a z-shaped chamber shield. The lower chamber shield **121** rests on the bottom of the chamber body **120** on an annular boss **177** formed on the bottom of the lower chamber shield **121**. The quartz prevents radiative coupling between the bottom of the heater pedestal **152** and the chamber body **120**. The annular boss **177** minimizes conductive heat transfer to the chamber body **120**. The lower chamber shield **121** includes an inwardly extending bottom lip joined to a conically shaped upper portion conforming to the inner wall of chamber body **120**, according to some embodiments.

[0026] A remote plasma source **141** is coupled to the processing chamber **150** through a gas port **132** to supply reactive plasma from the remote plasma source **141** through the plurality of holes **109** in the showerhead **156** to the processing chamber **150** to the substrate surface. It is noted that the remote plasma source **141** may be coupled to the processing chamber **150** in any suitable position to supply a reactive remote plasma source to the substrate surface as needed. Suitable gases that may be supplied to the remote plasma source **141** to be dissociated and further delivered to the substrate surface include H.sub.2, N.sub.2, Ar, He, ammonia (NH.sub.3), combinations thereof, and the like.

[0027] The control unit **180** is configured to control the various components of the processing chamber **150**. The control unit can be one of any form of a general purpose computer processor that can be used in an industrial setting for controlling various chambers and sub-processors. As shown, the control unit **180** includes one or more central processing units (CPUs) **182**, support circuitry **184**, and memory **186**. The CPUs can be configured within the control unit **180** to operate in sequence with one another or to operate in parallel with one another, and may communicate with one another via a message passing interface (not shown). The CPUs **182** can use any suitable memory **186**, such as random access memory, read only memory, floppy disk drive, compact disc drive, hard disk, or any other form of digital storage, local or remote. Various support circuits may be coupled to the CPUs **182** for supporting the processing chamber **150**. The control unit **180** can be coupled to another controller that is located adjacent individual chamber components. Bi-directional communications between the control unit **180** and various other components of the processing chamber **150** are handled through numerous signal cables collectively referred to as signal buses. The control unit **180** can also be configured to control various chambers of a multi-chamber processing system **200** as described below in FIG. **2**.

[0028] FIG. **1**B illustrates a processing chamber **1000**, according to at least one embodiment. The processing chamber **1000** is configured to sputter material onto a substrate. The processing chamber **1000** is a physical vapor deposition (PVD) chamber, capable of depositing, for example, ruthenium, titanium, cobalt, aluminum oxide, aluminum, copper, tantalum, tantalum nitride, tungsten, or tungsten nitride on a substrate, according to at least one embodiment. Examples of suitable PVD chambers include the ALPS® Plus and SIP ENCORE® PVD processing chambers, both commercially available from Applied Materials, Inc., Santa Clara, of Calif. It is contemplated that processing chambers available from other manufactures may also be utilized to perform the embodiments described herein. As shown, the processing chamber **1000** includes a target **1042**, a substrate support pedestal **1052**, a magnetron **1070**, a process kit **1040**, a control unit **1001**, and a body **1020**.

[0029] The target **1042** is configured to deposit material on the interconnect structures **400** (FIG. **4**) formed on a substrate **153**. As shown, the target **1042** has a sputtering surface **1045**. The target **1042** is supported by a grounded adapter **1044** through a dielectric isolator **1046**. The target **1042** includes the material to be deposited on the substrate **1054** surface during sputtering, and can include copper for depositing as a seed layer in high aspect ratio features formed in the interconnect structures **400**. The target **1042** also includes a bonded composite of a metallic surface layer of sputterable material, such as copper, and a backing layer of a structural material, such as aluminum, according to at least one embodiment. The target **1042** can also include other materials, such as Ru, Co, Ta, Nb, V, Cu, or manganese (Mn).

[0030] The substrate support pedestal **1052** is configured to support a substrate **153**. As shown, the substrate support pedestal **1052** has a peripheral edge **1053**. The substrate support pedestal **1052** can be located within a grounded chamber wall **1050**. The substrate support pedestal **1052** supports the substrate **153** having high aspect ratio features to be sputter coated, the bottoms of which are in planar opposition to a principal surface of the target **1042**. The substrate support pedestal **1052** has a planar substrate-receiving surface disposed generally parallel to the sputtering surface of the target **1042**. The substrate support pedestal **1052** can be vertically movable through a bellows **1058** connected to a bottom wall **1060** of the processing chamber **1000** to allow the interconnect structures **400** to be transferred onto the substrate support pedestal **1052** through a load lock valve (not shown) in a lower portion of the processing chamber **1000**. The substrate support pedestal **1052** can then be raised to a deposition position, as illustrated in FIG. **1**B.

[0031] Processing gas can be supplied from a gas source **1062** through a mass flow controller **1064** into the lower portion of the processing chamber **1000**. A direct current (DC) power source **1048**, coupled to the processing chamber **1000**, is used to apply a negative voltage or bias to the target **1042**, according to at least one embodiment. A radio frequency (RF) power source **1056** can be coupled to the substrate support pedestal **1052** to induce a DC self-bias on the substrate **153**. The substrate support pedestal **1052** is grounded, according to at least one embodiment. The substrate support pedestal **1052** is electrically floated, according to at least one embodiment.

[0032] The magnetron **1070** is configured to provide a magnetic field to the target **1042** to sputter material of the target onto the substrate **153** below. As shown, the magnetron **1070** includes a plurality of magnets **1072**, a base plate **1074**, and a shaft **1076**. The magnetron **1070** is positioned above the target **1042**. The plurality of magnets **1072** is supported by the base plate **1074** connected to the shaft **1076**, which may be axially aligned with the central axis of the processing chamber **1000** and the substrate **153**. The plurality of magnets **1072** are aligned in a kidney-shaped pattern, according to at least one embodiment. The plurality of magnets **1072** produce a magnetic field within the processing chamber **1000** near the front face of the target **1042** to generate plasma, such that a significant flux of ions strike the target **1042**, causing sputter emission of target material. The plurality of magnets **1072** can be rotated about the shaft **1076** to increase uniformity of the magnetic field across the surface of the target **1042**. The magnetron **1070** can be a small magnet magnetron. The magnets **1072** can be both rotated and moved reciprocally in a linear direction substantially parallel to the face of the target **1042** to produce a spiral motion. The magnets **1072** can be rotated about both a central axis and an independently-controlled secondary axis to control both their radial and angular positions.

[0033] The process kit **1040** is configured to protect various components of the processing chamber **1000** from unwanted sputtering from the target **1042**. As shown, the process kit **1040** includes a lower shield **1080**, a one-piece upper shield **1086**, and an optional collimator **1010**. The lower shield **1080** has a support flange supported by and electrically coupled to the chamber wall **1050**. The lower shield **1080** can include multiple pieces, or the lower shield can be one piece. The lower shield **1080** can be grounded. The upper shield **1086** is supported by and electrically coupled to a flange of the adapter **1044**. The upper shield **1086** and the lower shield **1080** are electrically coupled, as are the adapter **1044** and the chamber wall **1050**. Both the upper shield **1086** and the lower shield **1080** can include stainless steel. The processing chamber **1000** can include a middle shield (not shown) coupled to the upper shield **1086**. In one embodiment, the upper shield **1086** and the lower shield **1080** are electrically floating within the processing chamber **1000**. In one embodiment, the upper shield **1086** and the lower shield **1080** are coupled to an electrical power source.
[0034] In one embodiment, the upper shield **1086** has an upper portion that closely fits an annular side recess of the target **1042** with a narrow gap **1085** between the upper shield **1086** and the target **1042**, which is sufficiently narrow to prevent plasma from penetrating and sputter coating the dielectric isolator **1046**. The upper shield **1086** may also include a downwardly projecting tip **1090**, which covers the interface between the lower shield **1080** and the upper shield **1086**, preventing them from being bonded by sputter deposited material.
[0035] The lower shield **1080** extends downwardly into a cylindrical outer band **1096**, which generally extends along the chamber wall **1050** to below the top surface of the substrate support pedestal **1052**. The lower shield **1080** has a base plate **1098** extending radially inward from the cylindrical outer band **1096**. The base plate **1098** includes an upwardly extending cylindrical inner band **1003** surrounding the perimeter of the substrate support pedestal **1052**. A cover ring **1002** rests on the top of the cylindrical inner band **1003** when the substrate support pedestal **1052** is in a lower/loading position and rests on the outer periphery of the substrate support pedestal **1052** when the pedestal is in an upper/deposition position to protect the substrate support pedestal **1052** from sputter deposition.
[0036] The lower shield **1080** encircles the sputtering surface **1045** of the target **1042** that faces the substrate support pedestal **1052** and also encircles a peripheral wall of the substrate support pedestal **1052**. The lower shield **1080** covers and shadows the chamber wall **1050** of the processing chamber **1000** to reduce deposition of sputtering deposits originating from the sputtering surface **1045** of the target **1042** onto the components and surfaces behind the lower shield **1080**. For example, the lower shield **1080** can protect the surfaces of the substrate support pedestal **1052**, portions of the substrate **153**, the chamber wall **1050**, and the bottom wall **1060** of the processing chamber **1000**.
[0037] The collimator **1010** is configured to trap metal ions and neutrals that are emitted from the target **1042** at angles exceeding a selected angle, near normal to the substrate **153**. Directional sputtering is achieved by positioning the collimator **1010** between the target **1042** and the substrate support pedestal **1052**. The collimator **1010** is mechanically and electrically coupled to the upper shield **1086**. The collimator **1010** is attached to the upper shield **1086** by a plurality of radial brackets **1011**. In one embodiment, the collimator **1010** is coupled to a middle shield (not shown), positioned lower in the processing chamber **1000**. In one embodiment, the collimator **1010** is integral to the upper shield **1086**. In one embodiment, the collimator **1010** is welded to the upper shield **1086**. In one embodiment, the collimator **1100** is electrically floating within the processing chamber **1000**. In one embodiment, the collimator **1010** is coupled to an electrical power source. The collimator **1010** includes a plurality of apertures (omitted from FIG. **1**) to direct gas and/or material flux within the chamber. The operation of the processing chamber **1000** and the function of the collimator **1010** are similar regardless of the exact shape of the radial decreasing aspect ratio of the collimator **1010**.

[0038] The control unit **1001** is configured to control the various components of the processing chamber **1000**. The control unit **1001** can be one of any form of a general purpose computer processor that can be used in an industrial setting for controlling various chambers and sub-processors. As shown, the control unit **1001** includes a central processing unit (CPU) **1082**, support circuitry **1084**, and memory **1088**. The CPU **1082** can use any suitable memory **1088**, such as random access memory, read only memory, floppy disk drive, compact disc drive, hard disk, or any other form of digital storage, local or remote. Various support circuits may be coupled to the CPU **1082** for supporting the processing chamber **1000**. The control unit **1001** can be coupled to another controller that is located adjacent individual chamber components. Bi-directional communications between the control unit **1001** and various other components of the processing chamber **1000** are handled through numerous signal cables collectively referred to as signal buses. The control unit **1001** can also be configured to control various chambers of a multi-chamber processing system **200** as described below in FIG. **2**. The control unit **1001** can also control the processing chamber **150** described above.

[0039] In one embodiment, the control unit **1001** provides signals to position the substrate **153** on the substrate support pedestal **1052** and generate plasma in the processing chamber **1000**. The control unit **1001** sends signals to apply a voltage via the DC power source **1048** to bias the target **1042** and to excite processing gas, such as argon, into plasma. The control unit **1001** further provides signals to cause the RF power source **1056** to DC self-bias the substrate support pedestal **1052**. The DC self-bias attracts positively charged ions created in the plasma deeply into high aspect ratio vias and trenches on the surface of the substrate **153**.

[0040] Additionally, in order to provide even greater coverage of sputter deposited material onto the bottom and side walls of high aspect ratio features, material sputter deposited onto the field and bottom regions of features may be sputter etched. In one embodiment, the control unit **1001** applies a high bias to the substrate support pedestal **1052** such that the target **1042** ions etch film already deposited on the substrate **1054**. As a result, the field deposition rate onto the substrate **1054** is reduced, and the sputtered material re-deposits on either the side walls or bottom of the high aspect ratio features. In one embodiment, the control unit **1001** applies high and low bias to the substrate support pedestal **1052** in a pulsing, or alternating fashion such that the process becomes a pulsing deposit/etch process. In one embodiment, the collimator **1010** cells specifically located below magnets **1072** direct the majority of the deposition material toward the substrate **1054**. Therefore, at any particular time, material in one region of the substrate **153** is deposited, while material already deposited in another region of the substrate **153** is etched. In one non-limiting example, a Co material can be etched while a Ru material is deposited (e.g., at same time) without increasing liner overhang on top part of the interconnect **400**.

[0041] In one embodiment, to provide even greater coverage of sputter deposited material onto the side walls of high aspect ratio features, material sputter deposited onto the bottom of the features may be sputter etched using secondary plasma, such as argon plasma, generated in a region of the processing chamber **1000** near the substrate **153**. The processing chamber **1000** includes a coil **1041** attached to the lower shield **1080** by a plurality of coil standoffs **1043**, which electrically insulate the coil **1041** from the lower shield **1080**. The control unit **1001** sends signals to apply RF power through the shield **1080** to the coil **1041** via feedthrough standoffs (not shown). In one embodiment, the coil **1041** inductively couples RF energy into the interior of the processing chamber **1000** to ionize precursor gas, such as argon, to maintain secondary plasma near the substrate **1054**. The secondary plasma resputters a deposition layer from the bottom of a high aspect ratio feature and redeposits the material onto the side walls of the feature. The coil **1041** can include any metal, including Cu, Co, Ru, Ta, V, Nb, or combinations thereof.

[0042] In addition, in some embodiments, the processing chamber **1000** includes one or more lamps **1099**. The one or more lamps **1099** provide heat to the substrate **153**. In these embodiments, the processing chamber is identified by the index **1000′**.

[0043] FIG. **2** illustrates a schematic top view of a multi-chamber processing system **200**, according to at least one embodiment. The multi-chamber processing system is configured to perform deposition processes as disclosed herein having a processing chamber **150**, **1000**, **1000′** as described above in reference to FIGS. **1**A-**1**B, integrated therewith. As shown, the multi-chamber processing system **200** includes load lock chambers **202**, **204**, processing chambers **1000**, **1000′**, **150**, **216**, **232**, **234**, **236**, **238**, robots **210**, **230**, and transfer chambers **222**, **224**.

[0044] The load lock chambers **202**, **204** are configured to transfer substrate **153** into and out of the multi-chamber processing system **200**. Generally, the multi-chamber processing system **200** is maintained under vacuum and the load lock chambers **202** and **204** can be "pumped down" to introduce substrate **153** introduced into the multi-chamber processing system **200**. The first robot **210** transfers the substrate **153** between the load lock chambers **202** and **204**, and a first set of one or more processing chambers **1000**, **1000′**, **216**, and **150**. Each processing chamber **1000**, **1000′**, **216**, and **150** is configured to be at least one of a substrate deposition process, such as cyclical layer deposition (CLD), atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), etch, degas, pre-cleaning orientation, anneal, electrochemical plating (ECP), and other substrate processes. Furthermore, one of the processing chambers **1000**, **1000′**, **216**, and **150** can be configured to perform a pre-clean process prior to performing a deposition process or a thermal annealing process on the substrate **153**. The position of the processing chamber **150** relative to the other processing chambers **1000**, **1000′**, and **216** is for illustration, and the position of the processing chamber **150** can optionally be switched with any one of the processing chambers **1000**, **1000′**, and **216** if desired.

[0045] The first robot **210** can also transfer the substrate **153** to/from one or more transfer chambers **222** and **224**. The transfer chambers **222** and **224** can be used to maintain ultrahigh vacuum conditions while allowing substrate **153** to be transferred within the multi-chamber processing system **200**. A second robot **230** can transfer the substrate **153** between the transfer chambers **222** and **224** and a second set of one or more processing chambers **232**, **234**, **236** and **238**. Similar to the processing chambers **1000**, **1000′**, **216**, and **150**, the processing chambers **232**, **234**, **236**, and **238** can be outfitted to perform a variety of substrate processing operations including the dry etch processes described herein in addition to cyclical layer deposition (CLD), atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), etch, pre-clean, degas, and orientation, for example. Any of the processing chambers **1000**, **1000′**, **216**, **232**, **234**, **236**, and **238** can be removed from the multi-chamber processing system **200** if not necessary for a particular process to be performed by the multi-chamber processing system **200**. After the preclean, deposition and/or a thermal annealing process is performed in the processing chamber **150**, the substrate may further be transferred to any of the processing chambers **1000**, **1000′**, **216**, **232**, **234**, **236**, and **238** of the system **200** to perform other process as needed.

[0046] FIG. **3** is a flow diagram of method **300** operations for depositing a plurality of layers in a device structure **409** (FIG. **4**A) that includes an interconnect structure **400** formed on a substrate **153**, according to at least one embodiment. FIGS. **4**A-**4**D illustrate a schematic side view of a device structure **409** during various operations of the method **300** of FIG. **3**, according to at least one embodiment. The plurality of layers can include any of the layers deposited as described in the method **300** below. Although the method **300** operations are described in conjunction with FIGS. **3** and **4**A-**4**D, persons skilled in the art will understand that any system configured to perform the method operations, in any order, falls within the scope of the embodiments described herein. Any of the method **300** operations can be performed in any of the chambers of the multi-chamber processing system **200**, such as processing chamber **150**, **1000**, **1000′**, or in any other suitable chamber. In addition, different operations of the method **300** can be performed in different chambers within the multi-chamber processing system **200**, or in any other suitable chamber.

Deposition Process Sequence

[0047] FIG. **3** depicts a process flow diagram of a method **300** according to one or more embodiments of the disclosure. FIGS. **4**A-**4**D illustrate cross-sectional schematic views of an interconnect structure **400** during a process method **300** in accordance with one or more embodiments of the disclosure. It should be understood that FIG. **4**A-**4**D illustrate only partial schematic views of interconnect structure **400**, and the interconnect structure may contain any number of features and additional materials having aspects illustrated in the figures. It should also be noted that although the method **300** illustrated in FIG. **3** is described sequentially, other process sequences that include one or more operations that have been omitted and/or added, and/or have been rearranged in another desirable order, fall within the scope of the embodiments of the disclosure provided herein.

[0048] The interconnect structure **400** may include a first portion (“L1”) **402** and a second portion (“L2”) **404**. As illustrated in FIG. **4**A, the first portion **402** may be disposed substantially below second portion **404**, with a bottom surface **405** of second portion being coupled to first portion **402**. The bottom surface **405** of second portion **404** may include vertical surfaces **407** associated with a trench **401** and may extend below some portions of a top surface **403** of first portion **402**. The first portion **402** may include a dielectric layer **406**, a barrier layer **408**, a liner layer **410**, and interconnect layer **412**, and a etch stop layer **414**. The dielectric layer **406** may be formed of organosilicate glass (SiCOH), silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon carbide (SiC), aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), or any other dielectric material. The barrier layer **408** may be formed of tantalum, tantalum nitride, ruthenium doped tantalum nitride, or any other suitable material. The liner layer **410** may be formed of cobalt, ruthenium, titanium, tantalum, or any other suitable material. The interconnect layer **412**, which may be referred to as an m.sub.1 or an m.sub.0 layer, may be formed of molybdenum, tungsten, ruthenium, cobalt, copper, or any other suitable material. The etch stop layer **414** may be formed of silicon nitride, carbon, aluminum oxide (Al.sub.2O.sub.3), or any other suitable material. The second portion **404** may include a dielectric layer **416**, a barrier layer **418**, and a liner layer **420**. The dielectric layer **416** may be formed of organosilicate glass (SiCOH), silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon carbide (SiC), aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), or any other low-k dielectric material. The barrier layer **418** may be formed of tantalum, tantalum nitride, or any other suitable material. The liner layer **420** may be formed of cobalt, ruthenium, titanium, tantalum, or any other suitable material.

[0049] In some embodiments, at operation **302** of method **300**, the liner layer **420** is deposited over the already formed barrier layer **418** and the patterned dielectric layer **416**. The liner layer **420** and barrier layer **418** can be formed by use of a PVD, CVD, ALD, plasma enhanced CVD (PECVD), plasma enhanced ALD (PEALD), or other useful technique. FIG. **4**A illustrates an interconnect structure **400** after operation **302** has been performed. In FIG. **4**A, the liner layer **420** is disposed over the surface of the barrier layer **418** that is formed on the dielectric layer **414** disposed over the etch stop layer **414**. The liner layer **420**, the barrier layer **418**, the dielectric layer **416**, and the etch stop layer **414** each form part of the trench **401** formed in the dielectric layer **416**. The dielectric layer **416** is typically disposed on the etch stop layer **414**. The etch stop layer **414** is disposed on a dielectric layer **406** disposed within an underlying contact or interconnect structure (e.g., the first portion **402**). The dielectric layer **406** may be formed of a dielectric material similar to the dielectric material described above in conjunction with the dielectric layer **416**.

[0050] In some embodiments, during operation **302**, the liner layer **420** is a cobalt containing liner. The liner layer **420** may have a thickness between about 0.1 nm and about 5 nm, such as about 0.2 nm to about 3 nm, such as about 0.5 nm to about 2 nm, such that the liner layer **420** is formed of a conformal layer covering the surface of the barrier layer **418**. This thickness may be a minimum thickness to allow for copper reflow, which is discussed below with respect to operation **306**. The liner layer **420** may be a pure Co layer that is deposited using an ALD, CVD or PECVD process, such as those described above with respect to FIG. **2**.

[0051] In some embodiments, the barrier layer **418** may be formed with a thickness of between about 0.2 nm and about 2 nm, such as about 0.5 nm, to about 1.5 nm, such as about 1 nm. The barrier layer **418** may include one or more layers that include a ruthenium (Ru), tantalum (Ta), tantalum nitride (TaN), titanium (Ti), or titanium nitride (TiN) layer, and combinations thereof, that is deposited using PVD, ALD, CVD or PECVD processes, such as those described above with respect to FIG. **2**.

[0052] In some embodiments, method **300** includes depositing a second liner layer **422** at operation **304**. FIG. **4**B illustrates an interconnect structure **400** after operation **304** has been performed. In some cases, operation **304** may be performed at or near the same time as operation **302**. In FIG. **4**B, the second liner layer **422** is disposed at the same time as the liner layer **420**, over the surface of the barrier layer **418** that is formed on the dielectric layer **416**, disposed over the etch stop layer **414**. The second liner layer **422**, the liner layer **420**, the barrier layer **418**, the dielectric layer **416**, and the etch stop layer **414** include the trench **401** formed therein.

[0053] In some embodiments, the second liner layer **422** may be formed of ruthenium and the liner layer **420** may be formed of cobalt. The second liner layer **422** may have a thickness between about 0.1 nm and about 5 nm, such as about 0.2 nm to about 3 nm, such as about 2 nm. While the liner layer **420** is deposited according to operation **302**, the second liner layer **422** may be deposited using a PVD processes, such as those described above with respect to FIG. **1**. In some cases, the PVD process is implemented with little or no bias to encourage deposition of the second liner layer **422** with a gradational deposition pattern. Accordingly, the second liner layer **422** is deposited according to a concentration gradient of ruthenium, with more ruthenium material deposited towards the top portion of the trench **401**, and less ruthenium material deposited towards the bottom portion of the trench **401**, as noted by the change in shading of the second liner layer **422**. Some ruthenium material may be deposited on the liner layer **420** at the bottom of the trench **401**, but a greater concentration of the ruthenium material will reside at the field and upper portion of the trench **401**. It is believed that during the process of depositing the second liner layer **422** the ruthenium deposition will etch at least a portion of the liner layer **420** as it is deposited and form a combined liner layer that includes cobalt and ruthenium, such that liner overhang is not increased towards the top part of the trench **401** and the concentration of the ruthenium material in the combined liner layer is increased. As a result a lower portion of the combined layer includes a ruthenium doped cobalt layer. The resulting ruthenium concentration increases from a lower portion of the trench **401** to an upper portion of the trench **401**. Etching may be performed as part of the PVD processes described above, and may be formed by a physical bombardment process and/or a chemical reaction due to the exposure of the cobalt layer found in the liner layer **420** to the deposited ruthenium material. The second liner layer **422** may be considered a sacrificial layer, which may improve gap fill of metal within the trench **401**, while reducing overhang and void space. The sacrificial layer may be substantially removed during operation **308**.

[0054] In some embodiments, method **300** includes depositing a copper layer **424** at operation **306**. FIG. **4**C illustrates an interconnect structure **400** after operation **306** has been performed. In FIG. **4**C, the copper layer **424** is disposed over the second liner layer **422** and the liner layer **420** to substantially fill the trench **401** on the interconnect structures **400**. The copper layer **424** may be deposited using a metal deposition process, such as a PVD deposition process. Operation **306** also includes a reflow operation, which can include only partially reflowing the deposited copper layer **424**. The reflow process can include heating the substrate **153**, which includes the deposited copper layer **424**, to a reflow temperature greater than 150° C., such as between about 200° C. and about 600° C., such as about 400° C. in the same chamber as the copper deposition process was performed or in a separate processing chamber within the processing system. The reflow process can include depositing the copper layer **424** while the substrate **153** is heated to the reflow temperature. The reflow process can include depositing copper layer **424** before the substrate **153** is heated to the reflow temperature, then heating the substrate to the reflow temperature. The reflow process can include depositing the copper layer **424**. Thus, material forming the copper layer **424** may be heated to a reflow temperature before, during, or after depositing the copper layer.

[0055] It is believed that the formation of the second liner layer **422**, which includes a concentration gradient of ruthenium within the feature formed on the surface of the substrate **153**, improves the results of reflow of the copper layer **424** process by reducing the adhesion of the deposited copper layer to the liner layer **420** and encouraging diffusion of the copper into the lower portions of the trench **401**. This is believed to improve the step coverage of the deposited copper layer, reduce the amount of copper overhang after performing the copper deposition process, and minimize the chance of forming voids in the deposited copper containing feature after performing the reflow process as compared to conventional copper deposition, which is typically formed within a feature when the second liner layer **422** is not implemented prior to depositing the copper layer **424**. In some cases, this may enhance the strength and reliability of the liner layer **420**.

[0056] In some embodiments, method **300** includes the subsequent removal of the layers formed on the field region of the substrate during an operation **308**. FIG. **4**D illustrates an interconnect structure **400** after operation **308** has been performed. After the copper layer **424** is deposited in the trench **401** over the second liner layer **422** and the liner layer **420** at operation **306**, the second liner layer **422** of FIGS. **4**A-**4**C is substantially removed along with a portion **426** of the interconnect structures **400**. A new surface **428** of interconnect structures **400** is formed by removing a portion **426** of device structure **409**. Removing the portion **426** of device structure **409** may include the removal of a portion of the copper liner **424**, and the liner layer **420** and barrier layer **418** formed over the field region of the substrate.

[0057] In some embodiments, the second liner layer **422** is removed alongside of a portion **426** of the interconnect structures **400** using chemical-mechanical planarization (CMP) processes. CMP processes planarize the surface of the interconnect by applying precise downforce across the backside (not shown) of the interconnect structures **400** and pressing the top of the interconnect structures **400** against a rotating pad of special material that also contains a mixture of chemicals and abrasives.

[0058] The term “substrate” as used herein refers to a layer of material that serves as a basis for subsequent processing operations and includes a surface to be cleaned. The substrate may be a silicon based material or any suitable insulating materials or conductive materials as needed. The substrate may include a material such as crystalline silicon (e.g., Si<100> or Si<111>), silicon oxide, strained silicon, silicon germanium, doped or undoped polysilicon, doped or undoped silicon wafers and patterned or non-patterned wafers, silicon on insulator (SOI), carbon doped silicon oxides, silicon nitride, doped silicon, germanium, gallium arsenide, glass, or sapphire.

[0059] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method for forming an interconnect structure, comprising: depositing a ruthenium layer on a cobalt layer disposed within a feature formed on a substrate, the ruthenium layer having a ruthenium concentration that increases from a lower portion of the feature to an upper portion of the feature; and depositing a copper layer within the feature, wherein a material forming the copper layer is heated to a reflow temperature before, during, or after depositing the copper layer.

2. The method of claim 1, further comprising depositing the cobalt layer on a barrier layer, the barrier layer being disposed within the feature.

3. The method of claim 2, wherein the cobalt layer comprises a conformal layer covering a surface of the barrier layer.

4. The method of claim 2, wherein the cobalt layer has a thickness of about 0.5 nanometers (nm) to about 3 nm.

5. The method of claim 2, wherein the barrier layer comprises at least one of tantalum, tantalum nitride, and ruthenium.
6. The method of claim 2, wherein the cobalt layer is deposited using chemical layer deposition (CVD) or a plasma enhanced CVD (PECVD) process.
7. The method of claim 1, wherein the ruthenium layer has a thickness of about 0.5 nanometers (nm) to about 5 nm.
8. The method of claim 1, wherein depositing the ruthenium layer further comprises etching at least a portion of the cobalt layer formed in the feature.
9. The method of claim 1, wherein depositing the ruthenium layer is performed using a physical layer deposition (PVD) process, and a ruthenium concentration gradient in the cobalt layer is formed from the top of the feature to the bottom of the feature after depositing the ruthenium layer.
10. The method of claim 1, wherein depositing the ruthenium layer results in a ruthenium doped cobalt layer in at least a portion of the lower portion of the feature.
11. The method of claim 1, wherein depositing the ruthenium layer comprises etching the cobalt layer in the upper portion of the feature.
12. The method of claim 1, wherein depositing the copper layer comprises at least heating the substrate to a temperature greater than 200° C.
13. The method of claim 1, further comprising removing a portion of the interconnect structure, the portion of the interconnect structure comprising an upper portion of the ruthenium layer.
14. The method of claim 13, wherein removing the portion of the interconnect structure is performed using chemical-mechanical planarization (CMP).
15. A method for forming an interconnect structure, comprising: depositing a cobalt layer on a barrier layer disposed within a feature formed on a substrate; depositing a ruthenium layer on the cobalt layer disposed within the feature, the ruthenium layer having a ruthenium concentration that increases from a lower portion of the feature to an upper portion of the feature; and depositing a copper layer within the feature, wherein a material forming the copper layer is heated to a reflow temperature before, during, or after depositing the copper layer.
16. The method of claim 15, wherein depositing the ruthenium layer results in a ruthenium doped cobalt layer in the lower portion of the feature.
17. The method of claim 15, wherein depositing the ruthenium layer further comprises etching at least a portion of the cobalt layer.
18. A method for forming an interconnect structure, comprising: depositing a ruthenium layer on a cobalt layer disposed within a feature formed on a substrate, the ruthenium layer having a ruthenium concentration that increases from a lower portion of the feature to an upper portion of the feature; depositing a copper layer within the feature, wherein a material forming the copper layer is heated to a reflow temperature before, during, or after depositing the copper layer; and removing a top portion of the interconnect structure, the top portion of the interconnect structure comprising a first portion of the ruthenium layer.
19. The method of claim 18, wherein depositing the ruthenium layer results in a ruthenium doped cobalt layer in the lower portion of the feature.
20. The method of claim 18, wherein depositing the ruthenium layer further comprises etching the cobalt layer.